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### Key Points:

- We develop a framework that combines Nuclear magnetic resonance simulation and measurement to investigate pore coupling in unsaturated porous media
- Our simulation and measurement results reflect that the decrease in saturation weakens the degree of pore coupling
- Weakening pore coupling corresponds to more tortuous and eventually disconnected water-pathways due to reduced saturation

### Supporting Information:

Supporting Information may be found in the online version of this article.

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## How Does Fluid Exchange Between Pores in Unsaturated Porous Media?

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**Abstract** The inter-pore fluid exchange, driven by molecular diffusion, is referred to as pore coupling. The pore coupling affects subsurface pore-scale physicochemical dynamics. Monitoring pore coupling is essential for understanding numerous hydrological and ecological processes. Nuclear magnetic resonance (NMR) has effectively tracked pore coupling in saturated porous media. However, monitoring unsaturated pore coupling with NMR is challenging due to the complex pore network architecture and dynamic water-pathway connectivity. Based on X-ray computed tomography imaging, we develop a framework to simulate the NMR  $T_2$ -distribution and  $T_2$ -store- $T_2$  map of samples with distinct-sized pore environments at different water saturations, validated against measurements. NMR results, combined with visualized pore networks and phase distributions, demonstrate that reduced saturation induces increasingly tortuous and eventually disconnected water-pathways, thus restricting and ultimately blocking pore coupling. Our work enhances NMR applications in unsaturated media, providing accurate insights into the complex architecture and hydrodynamics in the pore network.

**Plain Language Summary** Fluid is widely retained within the pore network in shallow vadose zone soils, weathered bedrock, and deep oil and gas formations. Monitoring inter-pore fluid diffusion and exchange (pore coupling) through the pore network is crucial for managing groundwater resources, preserving ecosystems, and mitigating environmental hazards. Nuclear magnetic resonance (NMR) is a powerful geophysical method to investigate pore-scale fluid dynamics. Although NMR can provide detailed insights into pore coupling within saturated soils and rocks, its application in unsaturated subsurface materials remains challenging, even at the laboratory scale. To address this challenge, we developed a framework to integrate measurements and simulations to track the pore coupling and its NMR response under the unsaturated state. First, using X-ray micro computed tomography, we visualize detailed pore network structures and phase distributions to simulate NMR results validated against measurements, investigating how pore coupling affects unsaturated NMR responses. The results also show that as saturation decreases, weakening pore coupling tracked by NMR detects more tortuous and eventually disconnected water-pathways, enabling the evaluation of connectivity in unsaturated pore networks by applying NMR to monitor pore coupling. Our framework enhances our understanding of how NMR can be applied in tracking hydrodynamics within a partially saturated subsurface environment.

## 1. Introduction

Extensive pore networks exist throughout subsurface porous media, from shallow vadose zone soils to deep aquifers and hydrocarbon reservoirs, governing underground fluid migration and storage. Within these pore networks, fluid molecules continuously diffuse and exchange between pores (e.g., Anand & Hirasaki, 2007; Grunewald & Knight, 2009; Toumelin et al., 2003), a process known as diffusional pore coupling. This inter-pore exchange influences pore-environment behaviors, enhancing pore-scale heat redistribution (Koch et al., 2021), promoting solute mixing (e.g., Baqer & Chen, 2022; Sadeghi et al., 2003; Jiménez-Martínez et al., 2020; Perez et al., 2023), and affecting pore-scale fluid-rock interactions by modifying local concentration gradients. Consequently, pore coupling alters key hydrodynamic and thermodynamic processes and regulates reaction kinetics within the subsurface. These effects influence natural processes such as freeze-thaw dynamics in permafrost aggregates (Rooney et al., 2022) and contaminant migration in the vadose zone (Chen et al., 2023), as well as engineered applications such as recovery and management of hydrocarbon reservoirs (Li et al., 2024).

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Despite its importance, the mechanisms controlling pore coupling remain poorly understood. Previous experimental studies have visualized diffusion-driven transport and its effects on pore-scale fluid kinetics in millifluidic or microfluidic devices with simplified pore systems using imaging techniques such as scanning electrochemical microscopy or high-speed camera (e.g., Jiménez-Martínez et al., 2015; Liu et al., 2016; Zhang et al., 2025; Zhou et al., 2022). Numerical simulations have also been applied to model pore-scale diffusion reactions based on idealized pore network models (e.g., Collin et al., 2025; Li et al., 2017). However, these idealized frameworks lack the complexity of natural pore network architectures and cannot capture the dynamic connectivity of water-pathways in partially saturated materials, where solid–liquid–vapor interactions strongly influence molecular diffusion. In particular, how the degree of pore coupling changes with decreasing saturation, and how this change can be observed and quantified, remains unresolved.

Low-field nuclear magnetic resonance (NMR) provides a powerful, non-invasive way to investigate fluid behavior within porous media due to its sensitivity to hydrogen in fluid molecules (Coates et al., 1999; Kirkland & Codd, 2018; Prammer, 2004). NMR measures magnetization and relaxation times ( $T_1$  or  $T_2$ ) of protons after an applied magnetic field removal to estimate water content, pore size distribution, and hydraulic conductivity (Behroozmand et al., 2015). In multimodal pore systems, pore coupling drives inter-pore exchange of magnetized protons, as shown in Figures 1a and 1b. This exchange averages NMR relaxation behavior (Soto Bravo & Zhang, 2023), causing  $T_2$  peaks to shift or merge, which masks individual pore signatures and can make a multimodal system appear unimodal (Ramakrishnan et al., 1999). The 2D  $T_2$ -store- $T_2$  map captures this process through asymmetric off-diagonal peaks, which indicate magnetized proton exchange between pores and reflect both proton chemical shifts and inter-pore molecular exchange during mixing periods (Song, 2013; F. Zhang & Zhang, 2021). While NMR effectively tracks pore coupling in fully saturated materials (e.g., Anand & Hirasaki, 2007; Grunewald & Knight, 2009), variations in water-pathway geometry and connectivity complicate its application under unsaturated conditions. Accurately quantifying pore coupling and its NMR responses in unsaturated porous media is crucial to fully leverage NMR's potential for investigating pore-scale hydrodynamics and advancing understanding of subsurface physicochemical processes such as mineral dissolution and soil swelling.

To address this knowledge gap, we develop a framework combining NMR simulation and measurement with X-ray micro computed tomography ( $\mu$ CT) imaging to investigate pore coupling evolution in unsaturated multimodal-pore systems. Specifically, we: (a) integrate  $\mu$ CT imaging to visualize detailed pore network structures and phase distributions; (b) simulate NMR  $T_2$ -distributions and  $T_2$ -store- $T_2$  map at varying saturations and validate against laboratory measurements; (c) quantify how decreasing saturation weakens pore coupling by tracking increasingly tortuous and eventually disconnected water pathways. Our results substantially enhance NMR applications for characterizing unsaturated media and reveal how water-pathway connectivity governs pore coupling evolution, with implications for field-scale NMR investigations of subsurface hydrodynamics.

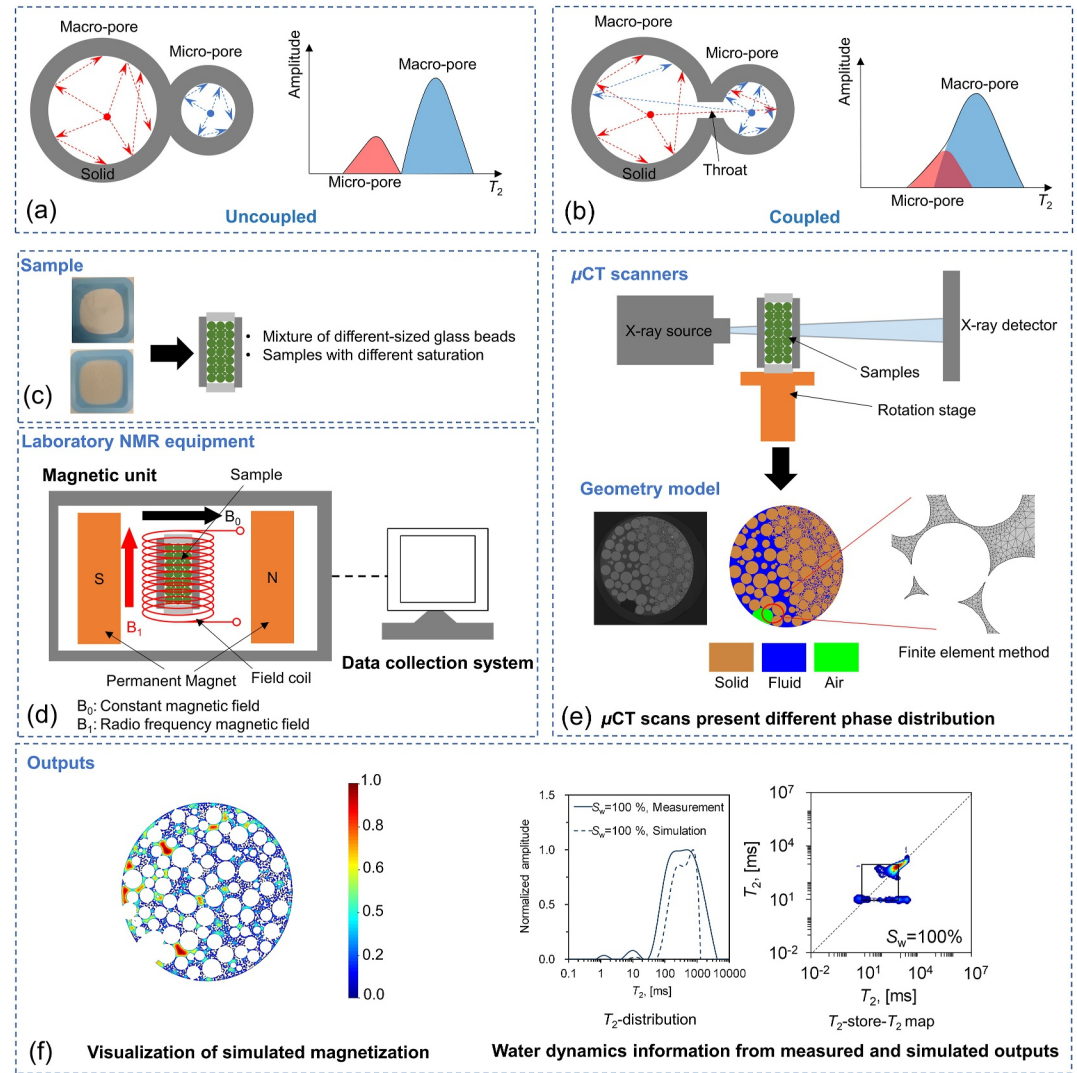
## 2. Experimental and Simulation Methods

### 2.1. NMR Theory and Relaxation Mechanisms

The spatiotemporal evolution of magnetization during NMR excitation and relaxation of protons in pore fluids is described by the Bloch-Torrey equation (Bloch, 1946; Torrey, 1956). The relaxation process manifests as transverse ( $T_2$ ) or longitudinal ( $T_1$ ) relaxation (Brownstein & Tarr, 1979), dominated by three mechanisms: bulk ( $T_{1B, 2B}$ ), surface ( $T_{1S, 2S}$ ), and diffusion relaxation ( $T_{2D}$ ). These combined relaxation rate can be expressed as:

$$\frac{1}{T_{1,2}} = \frac{1}{T_{1B,2B}} + \frac{1}{T_{1S,2S}} + \frac{1}{T_{2D}} \quad (1)$$

Bulk relaxation takes place spontaneously in the fluid and depends on fluid properties. Diffusion relaxation occurs in inhomogeneous magnetic fields and only affects  $T_2$ . Surface relaxation occurs in confined pore systems, and the relaxation rate of excited nuclei is enhanced by surface interactions at the fluid-solid interfaces, driven by the magnetic dipolar interactions between fluid spins and interfacial magnetic dipoles (Foley et al., 1996; Torrey, 1956). The active boundary condition is defined to reflect these molecular interactions at the solid-fluid boundary. In geologic media, the surface relaxation mechanism governs the relaxation process in the absence of magnetic field inhomogeneities. As a result, Equation 1 can be written as



**Figure 1.** Scenarios of (a) uncoupled and (b) coupled pore system and their respective relaxation time distributions; schematic diagram of (c) Sample preparation and (d) Nuclear magnetic resonance (NMR) experiment, (e) numerical simulation of NMR based on  $\mu$ CT imaging, and (f) NMR outputs.

$$\frac{1}{T_{1,2}} \approx \frac{1}{T_{1S,2S}} = \rho_s \frac{S}{V} \quad (2)$$

Here,  $S$  represents the surface area of water-filled pores,  $\rho_s$  is the surface relaxivity, and  $V$  is fluid-filled pore volume. Equation 2 reflects a well-perceived correlation between NMR relaxation time and pore size. In the unsaturated system, as air is the non-contributing passive volume to NMR responses (Mohnke & Klitzsch, 2010), the passive boundary condition is defined to qualify the magnetization at fluid-air interfaces. For complete details on Bloch-Torrey equation, boundary conditions, and derivations, refer to Section 1. Theory of NMR simulation and measurement of Text S1 in Supporting Information S1.

## 2.2. NMR Measurement Protocol

Multimodal pore samples were prepared by mixing glass beads of varying sizes and packing them into thin-wall polypropylene tubes (10 mm inner diameter and 20 mm length). Each sample at a specific water saturation  $S_w$  was placed into a cylindrical holder and loaded into the 12 MHz NMR core analyzer from Niumag MacroMR (Figure 1c). Here,  $S_w = V_w/V_{\text{pore}}$ , where  $V_w$  is water volume and  $V_{\text{pore}}$  is total pore volume.

Two types of NMR measurements were applied. First, we performed  $T_2$  relaxation measurements on samples at different saturations using the Carr-Purcell-Meiboom-Gill (CPMG) pulse sequence. Subsequently, we conducted 2D  $T_2$ -store- $T_2$  measurements for these samples using Transverse Relaxation Exchange (TReX) pulse sequence. The TReX sequence consists of two CPMG trains separated by a storage period, during which the magnetization is stored along the  $z$ -axis (Montrazi et al., 2018). During this storage time, magnetized protons diffuse between pore environments, enabling inter-pore magnetization exchange. A 600 ms storage time was selected to ensure sufficient inter-pore proton exchange and thereby accurately and quantitatively capture the degree of pore coupling in multimodal systems. The collected NMR signals were inverted to obtain  $T_2$ -distributions and 2D  $T_2$ -store- $T_2$  maps of samples at different saturations. Complete details on pulse sequence parameters, and inversion methods are provided in Section 2.  $T_2$ -distribution simulation and measurement and 3.  $T_2$ -store- $T_2$  map simulation and measurement of Text S1 in Supporting Information S1, while details on sample preparation procedures are provided in Sample for NMR measurement of Text S2 in Supporting Information S1.

### 2.3. $\mu$ CT Imaging and NMR Responses Simulation

The  $\mu$ CT imaging of samples was conducted using the HECTOR system (Masschaele et al., 2013) at the Ghent University Centre for X-ray Tomography (UGCT). The  $\mu$ CT setup is shown in Figure 1d. To achieve high-resolution imaging of phase distributions, smaller sample tubes (5 mm inner diameter and 20 mm length) were used compared to those for NMR measurements. Samples with different water saturations were prepared with 2.5% potassium iodide (KI) solution to enhance phase contrast. Sealed samples were scanned at 3  $\mu$ m voxel resolution, and three-phase segmentation (solid, fluid, and air) was performed on reconstructed 2D grayscale cross-sections along the vertical height (Figure 1b). These segmented images form a 3D volume visualizing pore network structures and phase distribution in samples.

To reduce computational cost and uncertainty, pore network models were constructed directly from 2D phase-segmented images and used to simulate NMR responses via a commercial finite element software. Images of each sample were clustered by vertical position, which is identified by similar phase distribution and pore geometry. Each cluster containing an equal number of images represents an equal-length height range. One representative image from each cluster was selected for pore network model construction. The fluid-occupied space from selected images was meshed, incorporating boundary conditions at fluid-air and fluid-solid interfaces, to solve the Bloch-Torrey equation for time-dependent magnetization distribution. Temporal variations in the external magnetic field in the Bloch-Torrey equation were accounted for to accurately simulate both CPMG and TReX pulse sequences. Magnetization signals from ten representative simulations were superimposed and inverted to  $T_2$ -distributions and 2D  $T_2$ -store- $T_2$  maps. Sample information refers to Sample for  $\mu$ CT imaging of Text S2 in Supporting Information S1. Image processing procedures (Step 1 to 3),  $\mu$ CT imaging analysis (Step 4 and 5), and simulation parameters specifications (e.g.,  $\rho_s$  calculated by Equation S11 and S12 in Supporting Information S1) (Step 4 and 5) are provided in Text S3 in Supporting Information S1.

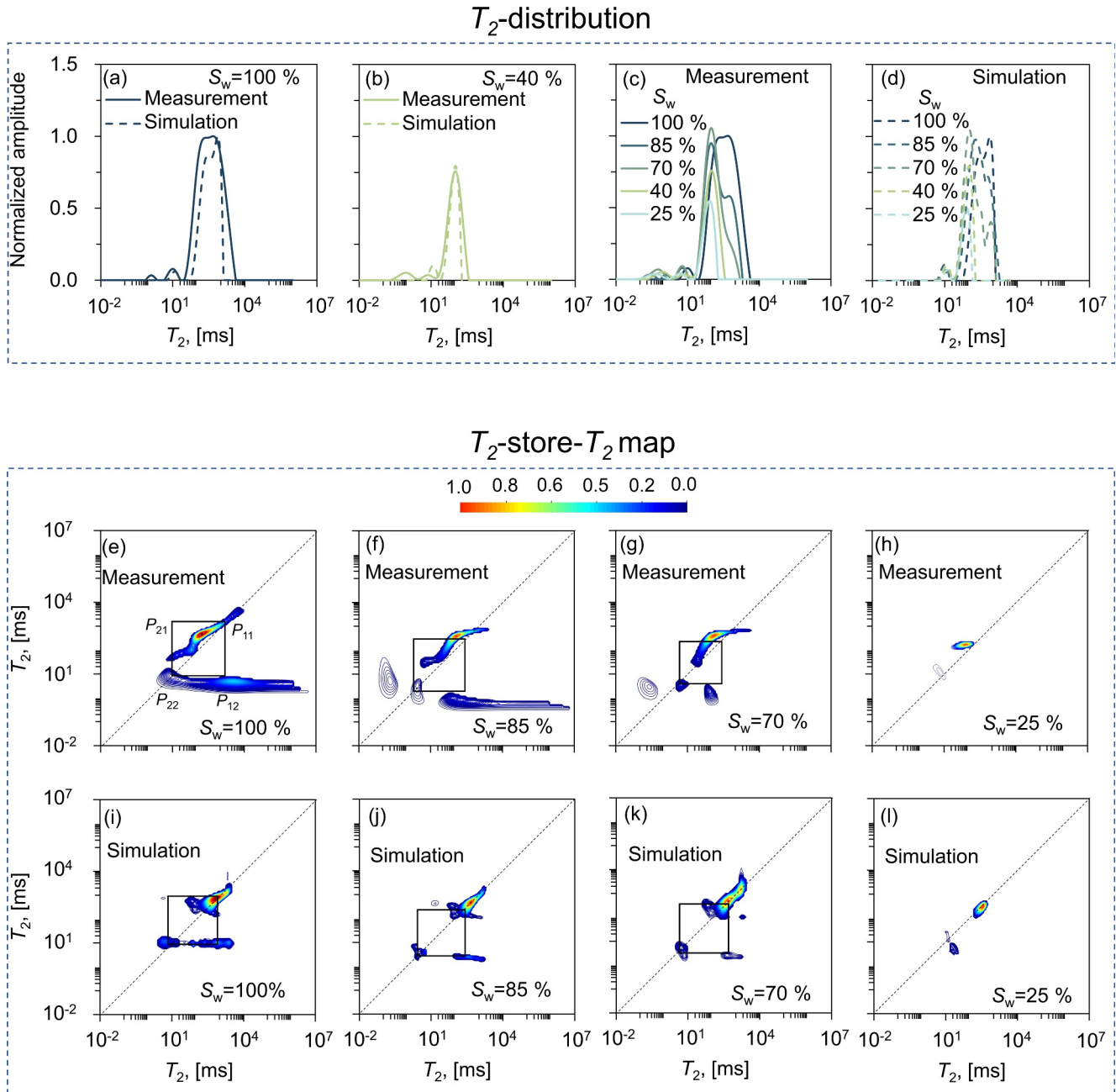
## 3. Results and Discussions

We combined  $\mu$ CT-based NMR simulation with experimental validation to quantify how pore coupling evolves with saturation. Both  $T_2$ -distribution and  $T_2$ -store- $T_2$  map reveal progressive weakening of pore coupling as saturation decreases, corresponding to increasingly tortuous and eventually disconnected water-pathways. This framework provides quantitative metrics for tracking pore coupling in unsaturated multimodal-pore networks.

### 3.1. The Pore Coupling Effect on $T_2$ -Distribution in Unsaturated Media

The  $T_2$ -distribution captures hydrogen relaxation signatures in the individual pore environment, with amplitudes reflecting their pore-water content. Inter-pore magnetization exchange distorts the  $T_2$ -distribution at different saturations, directly manifesting the pore coupling effect on NMR responses.

At full saturation ( $S_w = 100\%$ ), both simulated and measured  $T_2$ -distribution show multiple peaks corresponding to large and small pores (Figure 2a). With decreasing saturation, the amplitude of the longest-relaxation time peak decreases more significantly than that of the shorter-relaxation components, reflecting the preferential moisture reduction within the largest pores (Figures 2c and 2d). Notably, at  $S_w = 70\%$ , the shorter-relaxation component amplitude exceeds that at  $S_w = 100\%$  and 85%. This suggests that in the largest pores, unsaturated conditions enhance surface interactions, accelerating proton relaxation. As a result, some protons in large pores exhibit



**Figure 2.** Measured and simulated  $T_2$ -distribution and  $T_2$ -store- $T_2$  maps of samples at different saturation states: Comparison of simulated and measured  $T_2$ -distributions at (a)  $S_w = 100\%$ , (b)  $S_w = 40\%$ ; (c) Measured and (d) simulated  $T_2$ -distributions for samples at different saturations; Measured  $T_2$ -store- $T_2$  maps at (e)  $S_w = 100\%$ , (f)  $S_w = 85\%$ , (g)  $S_w = 70\%$ , and (h)  $S_w = 25\%$ ; Simulated  $T_2$ -store- $T_2$  maps at (i)  $S_w = 100\%$  (j)  $S_w = 85\%$  (k)  $S_w = 70\%$ , and (l)  $S_w = 25\%$ .

relaxation behavior similar to those in smaller pores, contributing to the increased short-relaxation signal (J. Zhou & Zhang, 2025). As saturation decreases to 40%, the longest-relaxation time peak (largest pores) disappears entirely, leaving only shorter-relaxation components (Figure 2b), suggesting that water is retained only in small pores.

At the  $S_w = 100\%$ , the two main relaxation time peaks nearly merged (Figures 2c and 2d). As saturation decreases, the peaks progressively separate, producing a clearer distinction between the long- and short-relaxation components at  $S_w = 85\%$  and 70%. This comparison implies that under saturated conditions, the strong pore coupling facilitates inter-pore magnetization exchange, resulting in merged relaxation time peaks. As saturation

decreases, pore coupling weakens, restricting inter-pore magnetization exchange and leading to clearer separation of the relaxation time peaks. While these observations directly reveal pore coupling evolution, the underlying mechanisms, such as the influence of pore channels' width, length, connectivity, and surface geochemistry (i.e. quantified by surface relaxivity ( $\rho_s$ )) require further analysis. We will examine what factors weaken pore coupling in multi-pore systems as saturation decreases in the following discussion.

This weakened pore coupling effect on NMR relaxation can also arise from variations in pore network geometry and connectivity characteristics, including reduced pore channel width (e.g., Anand & Hirasaki, 2007), longer lengths (e.g., Mohnke & Klitzsch, 2010), and fewer connecting channels (Chi & Heidari, 2015). Mineral surface geochemistry, as quantified by surface relaxivity ( $\rho_s$ ), can also regulate the inter-pore magnetization exchange. Higher  $\rho_s$  accelerates relaxation and thus weakens observable pore coupling effects (e.g., Grunewald & Knight, 2009). While our study focuses on geometric controls,  $\rho_s$  remains an important external factor that can influence NMR sensitivity to pore coupling. We further investigate the role of pore structure in governing coupling behavior as saturation decreases in the following discussion.

### 3.2. Quantitative Analysis of Unsaturated Pore Coupling From $T_2$ -Store- $T_2$ Map

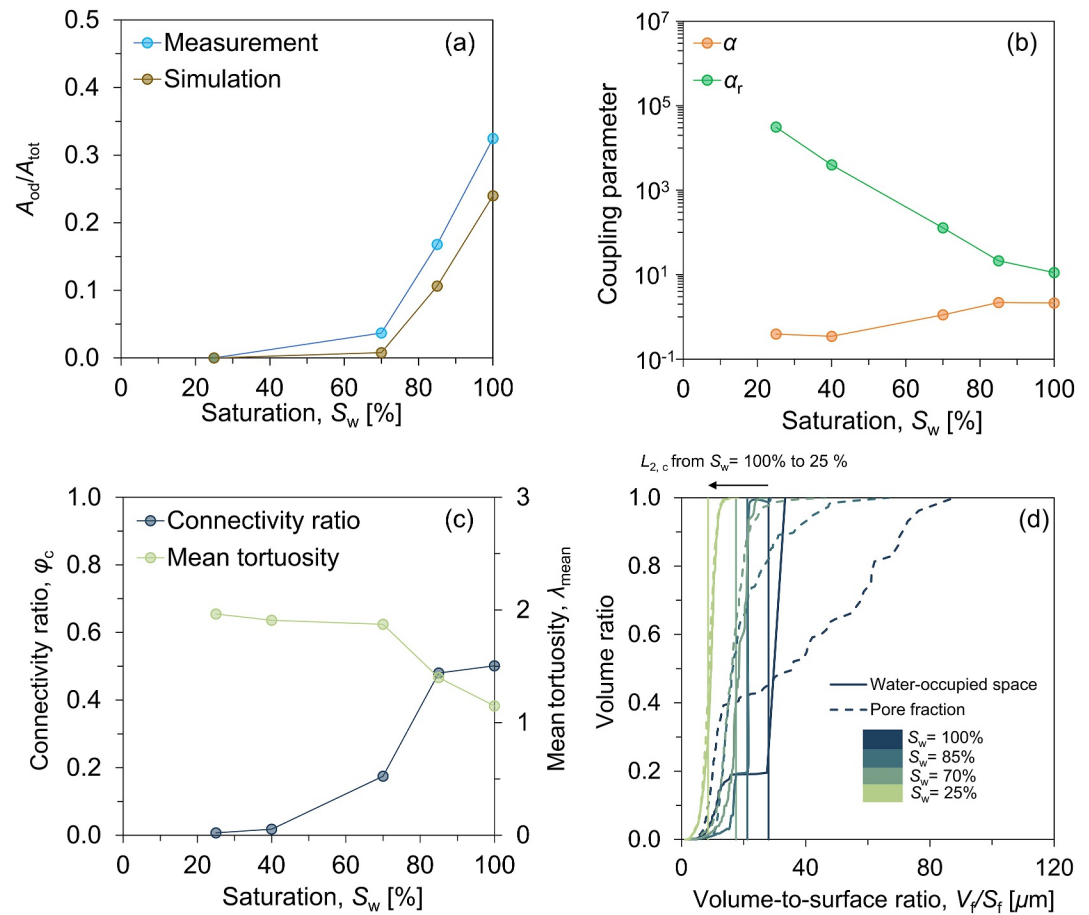
The  $T_2$ -store- $T_2$  map enables a qualitative assessment of inter-pore magnetization exchange, providing insight into the degree of pore coupling in unsaturated pore networks. In both measured (Figures 2e–2h) and simulated (Figures 2i–2l)  $T_2$ -store- $T_2$  maps, peak amplitudes are normalized by the maximum amplitude value in each map. The horizontal and vertical axes correspond to the  $T_2$  relaxation times of the same proton population before and after exchange. Thus, in Figure 2e, diagonal peaks  $P_{11, 22}$  respectively represent magnetized protons remaining within large and small pore environments, while off-diagonal peaks  $P_{12, 21}$  correspond to protons diffusing from large to small pores and small to large pores, respectively (Schwartz et al., 2013). Monitoring these off-diagonal peaks enables quantitative estimation of pore coupling under saturated conditions (e.g., Fleury & Soualem, 2009; Song et al., 2014; Washburn & Callaghan, 2006).

Generally, an equal number of exchanged protons between coupled pores produces symmetric off-diagonal peaks about the diagonal ( $T_2/T_2 = 1$ ). In this study, Figure 2 shows that only  $P_{12}$  appears in both simulated and measured  $T_2$ -store- $T_2$  maps, particularly at  $S_w = 100\%$  and  $85\%$ . This asymmetry is attributed to the stronger surface relaxation in the small pore population, accelerating NMR relaxation of protons. Consequently, even if sufficient and balanced inter-pore exchange occurs, the magnetization carried by protons may decay before migrating from small to large pores, suppressing the NMR signal corresponding to  $P_{21}$ . This work utilizes the saturation-dependent evolution of  $P_{12}$  to monitor magnetized proton migration from large to small pores, thereby quantitatively tracking the progressive weakening of pore coupling with decreasing saturation.

Both measured (Figures 2e–2h) and simulated (Figures 2i–2l)  $T_2$ -store- $T_2$  maps exhibit that the amplitude of off-diagonal peak  $P_{12}$  decreases with reducing sample saturation. This evolution implies a decreasing proportion of magnetized protons migrating from the large to small pore population. Figure 3a shows the total amplitude of the off-diagonal peaks  $A_{od}$  relative to the total peaks' amplitude  $A_{tot}$  ( $A_{od}/A_{tot}$ ) (Monteilhet et al., 2006), as a function of saturation. Both simulated and measured  $A_{od}/A_{tot}$  decline with decreasing saturation and approach zero at low saturation ( $S_w = 25\%$ ), coinciding with the disappearance of the off-diagonal peak. This quantitative trend reflects the reduced proportion of inter-pore magnetized proton exchange and demonstrates that  $T_2$ -store- $T_2$  maps provide insight into quantitatively tracking the weakening pore coupling in multi-pore systems as saturation decreases.

### 3.3. Comparison Between Simulation and Experimental Results

This work establishes pore-network models based on 3D  $\mu$ CT imaging of real porous media to simulate NMR responses. First, we simulate  $T_2$  distributions at different saturations to directly manifest the pore coupling effect on NMR responses. The simulation results are consistent with measurements, validating our framework. Compared with previous studies that only simulated  $T_2$ -store- $T_2$  maps for idealized saturated pore models (e.g., Fleury & Soualem, 2009), our framework simulates these maps to monitor the saturation-dependent pore coupling strength. While the simulated and measured  $T_2$ -distributions align well, some differences are observed in the peak shapes and relative positions of off-diagonal peaks in the  $T_2$ -store- $T_2$  maps. The observed discrepancies between the simulated and experimental  $T_2$ -store- $T_2$  maps mainly arise from two factors. For the simulation, NMR responses were obtained by superimposing results from 2D phase-segmented images at different



**Figure 3.** (a) The variation of  $A_{od}/A_{tot}$  during the desaturation process; The evolution of (b)  $\alpha_r$ , (c)  $\lambda_{mean}$ , and  $\phi_c$  during the desaturation process; (d) The cumulative distributions of  $V_f/S_f$  for connected pores and pore fractions at four saturations.

heights of the sample columns, which may potentially neglect vertical pore coupling. For the measurement, previous studies have presented that signal noise during data acquisition can cause deviations of off-diagonal peaks in inverted  $T_2$ -store- $T_2$  maps (via inverse Laplace transform) from the theoretical square positions formed with the diagonal peaks (Figures 2f and 2g) (e.g., Fleury & Soualem, 2009). This noise may also contribute to the trailing artifacts of the off-diagonal peaks observed in Figures 2c and 2f.

Despite these differences, the evolution of peak distribution and amplitude in simulated  $T_2$ -store- $T_2$  maps with varying saturations is broadly consistent with the experimental results. Specifically, the variation trend of  $A_{od}/A_{tot}$  with saturation derived from both simulation and measurement also matches well. Together, these results demonstrate that  $T_2$ -store- $T_2$  maps can be utilized to quantify pore coupling strength in pore networks with different structures and dynamic unsaturated pore-water distributions. Our simulation framework can predict NMR responses of unsaturated pore networks and provide insights into qualifying complex inter-pore water dynamics within subsurface porous media from NMR measurement data, thereby advancing our understanding of subsurface hydrological behaviors and processes.

### 3.4. Determinants of Pore Coupling Effect Under Unsaturated Conditions

In the following discussion, 3D  $\mu$ CT volumes of the samples are used to visualize the pore network structure and phase distribution. Based on the visualization, we elucidate factors responsible for the weakened pore coupling in multi-pore systems as saturation decreases, as revealed by the previous NMR simulation and measurement results. Multiple water-retained pores are interconnected by water-pathways, forming distinct water-occupied spaces within the unsaturated system. To demonstrate irregular water-pathways, the mean water-pathway tortuosity  $\lambda_{mean}$  is derived from skeletonized pathways in 3D  $\mu$ CT volumes. The effective diffusion coefficient is

defined as  $D_{\text{eff}} = D/\lambda_{\text{mean}}$ . The connectivity for water movement is quantified by the connectivity ratio ( $\varphi_c$ ), representing the probability that two randomly selected points belong to the same interconnected water-occupied region extracted from 3D  $\mu$ CT volumes. Details of  $\lambda_{\text{mean}}$  and  $\varphi_c$  are provided in the Step 5. Analysis of fluid distribution, fluid pathways tortuosity, and connectivity of Text S3 in Supporting Information S1. Surface relaxation occurs at the fluid-solid interface. The fluid volume ( $V_f^i$ ) and surface area ( $S_f^i$ ) of each connected space, and total values ( $V_f^{\text{tot}}$ ,  $S_f^{\text{tot}}$ ) of the entire system, are used to calculate the characteristic diffusion length  $L_2^i = V_f^i/S_f^i$ .  $L_2^i$  represents the average distance that surface-relaxing spins must diffuse throughout the space. Thereby quantifying the effective geometric size of the space. In the unsaturated system, Equation 2 can be expressed as  $T_{2,c} = V_f^{\text{tot}}/(\rho_s S_f^{\text{tot}})$  to define the characteristic relaxation time of system.

The coupling parameter  $\alpha = (L_2/\sqrt{DT_{2,c}})^2$ , proposed by Anand and Hirasaki (2007), quantifies pore coupling effect on NMR relaxation in saturated systems with fully connected and regular pore networks. We propose the re-normalized coupling parameter of one connected space, expressed as  $\alpha_r = (L_2/\varphi_c \sqrt{D_{\text{eff}} T_{2,c}})^2$ , to account for the effects of the surface geochemistry and irregular geometry of a partially connected water-pathway network in this space. The  $\alpha_r$  of the entire unsaturated system can be expressed as

$$\alpha_r = \sum_i \frac{V_f^i [L_2^i / (\varphi_c \sqrt{D_{\text{eff}} T_{2,c}})]^2}{\sum_i V_f^i} \quad (3)$$

to enable quantification of the pore coupling effect on NMR relaxation in the unsaturated pore network. Equation 3 indicates that magnetization decay in coupled-pore systems is governed by surface relaxation of spins and inter-pore exchange. When the diffusion length is much smaller than the diffusion distance within the characteristic relaxation time (i.e.,  $L_2^i \ll \varphi_c \sqrt{D_{\text{eff}} T_{2,c}}$ ), magnetized protons can diffuse between distinct pore environments before complete relaxation of their magnetization, leading to effective inter-pore magnetization exchange and thus strong pore coupling effect on NMR relaxation.

The  $\alpha_r$  increases with decreasing saturation (Figure 3b). The system remains in an intermediate coupling regime ( $1 < \alpha_r < 250$ ) for  $S_w > 40\%$  (where  $\alpha_r < 1$  represents total coupling in Anand & Hirasaki, 2007) and transitions to a decoupled state ( $\alpha_r > 250$ ) at  $S_w \leq 40\%$ . This trend aligns with the weakening observable pore coupling qualified by the evolution of  $A_{\text{od}}/A_{\text{tot}}$  with saturation. Figure 3b also presents that while the  $\alpha$  changes slightly, the  $\alpha_r$  varies significantly with saturation. These evolutions imply that variations in saturation primarily alter the water-pathway tortuosity and inter-pore connectivity in the irregular and partially connected pore network, which may potentially influence the degree of pore coupling. Figure 3c further illustrates the evolution of  $\lambda_{\text{mean}}$  and  $\varphi_c$  with saturation. As saturation decreases from 100% to 85%, the  $\lambda_{\text{mean}}$  increases significantly while the  $\varphi_c$  changes slightly. This implies that although the unsaturated pores remain well-connected at  $S_w = 85\%$ , the reduced inter-pore channel water induces narrower and more tortuous water-pathways. When saturation further decreases from  $S_w = 85\%$  to 70%, both  $\lambda_{\text{mean}}$  and  $\varphi_c$  change significantly, implying that some pathways become narrower and more tortuous, while others disconnect due to insufficient water. From  $S_w = 70\%$  to 25%,  $\varphi_c$  decreases sharply while  $\lambda_{\text{mean}}$  remains nearly constant, suggesting that further moisture reduction no longer alters water-pathway geometry but directly blocks inter-pore connectivity.

The comparison of cumulative distributions of  $V_f/S_f$  in interconnected water-occupied spaces (dashed line) and individual unsaturated pore fractions (dashed line) at four saturations (Figure 3d) also illustrates how varying saturation influences the inter-pore connectivity.  $L_{2,c} = V_f^{\text{tot}}/S_f^{\text{tot}}$  reflects the characteristic length for surface-relaxing spins to diffuse throughout the system under the total coupling state (Fleury & Soualem, 2009). At high saturation, the two curves differ significantly, with pore fractions exhibiting a much larger maximum  $V_f/S_f$ . This indicates that water-occupied spaces are formed by the connection of numerous unsaturated pores, with the effective geometry size of spaces represented by the average characteristic size of pores forming them. Especially, at  $S_w = 100\%$  and 85%, the water-occupied space curve displays a sharp rise near  $V_f/S_f = L_{2,c}$ , implying that most unsaturated pores are well-connected to form a main water-occupied space. As saturation decreases, two curves gradually evolve toward convergence. These pores no longer form a single interconnected space but segregate into isolated smaller pore clusters with limited connectivity, due to partial disconnection of water-

pathways. At low saturation, most pores become completely isolated, forming numerous independent water-occupied spaces.

The aforementioned evolution illustrates how saturation-induced variations in water-pathway geometry and inter-pore connectivity govern the pore coupling strength. As saturation decreases, reduced water content in inter-pore channels induces narrower and more tortuous pathways. This variation results in longer fluid diffusion paths, thereby restricting the inter-pore molecular exchange. At low saturation, insufficient water fails to maintain inter-pore connectivity. The interconnected water-occupied space progressively segregates into several smaller pore clusters with limited connectivity. Eventually, these clusters fragments into isolated and individual pore environments as most water-pathways become completely disconnected. Consequently, the inter-pore fluid diffusion is blocked, eliminating the pore coupling.

#### 4. Geophysical Implications and Broader Applications

Our results demonstrate that decreasing water saturation leads to progressively weaker pore coupling due to increased tortuosity and disconnection of water pathways. This process is directly reflected in the merging or broadening of  $T_2$  peaks and the loss of off-diagonal features in  $T_2$ -store- $T_2$  maps. This relationship has broad implications for interpreting NMR data in hydrological models that can be used for groundwater management and environmental remediation.

By quantifying coupling strength under variable saturation, NMR can inform parameterizations in dual-porosity or mobile-immobile domain models, especially in predicting the retention, release, or accessibility of water and solutes across pore scales. Furthermore, common NMR-derived permeability estimation models such as the Schlumberger Doll Research (Kenyon et al., 1988) and Timur-Coates (Coates et al., 1999; Timur, 1969) models are calibrated for saturated conditions and may significantly misestimate hydraulic conductivity in unsaturated vadose zones. While several previous studies have applied NMR-derived parameters to estimate subsurface hydraulic conductivity in unsaturated media (e.g., Legchenko et al., 2004; Legchenko & Vouillamoz, 2011; Vouillamoz et al., 2003), these approaches generally rely on averaged relaxation properties and do not explicitly account for pore coupling, which reflects saturation-dependent inter-pore connectivity. As a result, permeability estimates may be inaccurate under partially saturated conditions. The degree of pore coupling, quantified through  $A_{od}/A_{tot}$  and  $\alpha_r$ , provides correction factors for estimating hydraulic conductivity in partially saturated conditions. Field NMR instruments, including surface NMR and borehole NMR, detect  $T_2$ -distribution profiles along depth in the natural subsurface (e.g., Guérin, 2005; Jiang et al., 2020; Kirkland & Codd, 2018). Our findings clarify that vertical variations in  $T_2$  spectra might reflect pore coupling evolution due to saturation changes. This distinction enhances field data interpretation, particularly in heterogeneous formations where saturation gradients are strong.

Beyond NMR, other geophysical methods such as spectral induced polarization (SIP) are sensitive to pore-scale connectivity and diffusion-controlled charge transport (Tsukanov et al., 2021), the same processes quantified through our pore coupling framework. Integrating NMR measurements with SIP and other geophysical methods enables more robust characterization of subsurface fluid dynamics and reactive transport, improving predictive models of water-rock interactions in complex geologic environments.

#### 5. Conclusion

This study aims to clarify how NMR responds to the pore coupling in unsaturated porous media, and how variation in saturation governs the pore coupling strength. Specifically, this paper develops a framework combining simulation and measurement to investigate pore coupling and its NMR responses in unsaturated porous media and provides insights into investigating hydrodynamics within complex subsurface geologic environments through NMR observation.

Realistic pore-network models constructed from 3D  $\mu$ CT imaging are used to simulate the evolution of  $T_2$ -distribution and  $T_2$ -store- $T_2$  map from saturated to unsaturated states, validated against NMR measurement results. The evolution of  $T_2$ -distribution reveals that the decreased saturation weakens the pore coupling effect on NMR relaxation. This weakened pore coupling strength can be quantitatively tracked by  $T_2$ -store- $T_2$  maps.

Based on  $\mu$ CT imaging, we introduce the mean water-pathway tortuosity  $\lambda_{mean}$  and connectivity ratio  $\phi_c$  into the re-normalized coupling parameter  $\alpha_r$ . The  $\alpha_r$  quantitatively evaluated the attenuation of the degree of pore coupling effect on NMR relaxation with saturation decreasing. The evolution of  $\lambda_{mean}$  and  $\phi_c$ , combined with the

comparison of cumulative distribution curves of  $V_f/S_f$  for interconnected water-occupied space and individual pore fractions at different saturations, clarify the factors responsible for weakened pore coupling: from fully saturated state to high saturation condition, water-pathways become increasingly tortuous, as indicated by the rise in  $\lambda_{\text{mean}}$ , which restrict the inter-pore fluid exchange; as saturation further decreases, these tortuous pathways progressively disconnect, and the interconnected water-occupied space progressively segregates into numerous independent unsaturated pores, leading to a sharp decline in  $\phi_c$  and ultimately blocking pore coupling. This mechanism suggests that NMR tools can be used to monitor the degree of pore coupling, thereby enabling the evaluation of pore-scale connectivity and diffusion-controlled transport under unsaturated conditions.

## Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

## Data Availability Statement

All data of CT images, NMR responses simulation, and NMR measurement in the article are available at Science Data Bank, via J. Zhou (2025), with CC-BY 4.0.

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